Assignment 2 – Road Collision Analysis (Individual)

Overview

You will clean, prepare, and analyse the given **UK road collision data**. Your task is to identify and address data quality issues, combine the two provided files where appropriate, and produce a concise analysis with clearly supported conclusions.

Use **Pandas** and **NumPy** for all data handling. There is no need to visualize anything.

All data files are in the **datasets** folder. Work only with files in that folder as the input dataset for your work.

Notebook Requirement
You must complete this assignment in a **Jupyter Notebook (.ipynb)** file.
Only **.ipynb** submissions will be accepted; **.py** or other formats **will not be marked.**

Files Provided

1. **road_collision_2023.csv:** Collision-level information (one row per collision).

Column	Description	Example
collision_index	Unique collision identifier	2023170L30453
collision_year	Year of collision	2023
police_force	Police force area code	17
collision_severity	Severity (1 = Fatal, 2 = Serious, 3 = Slight)	3
number_of_vehicles	Number of vehicles involved	3
number_of_casualties	Number of casualties	1
speed_limit	Speed limit at location	60
weather_conditions	Weather conditions code	1
light_conditions	Lighting conditions code	1
urban_or_rural_area	Area type	2
collision_adjusted_severity_serious	Proportion of serious casualties (0–1 scale)	0.014

2. **road_vehicle_2023.csv:** Vehicle-level information (multiple rows per collision).

Column	Description	Example
collision_index	Matches collision file	2023481356437
vehicle_type	Type of vehicle	9
age_of_driver	Age of driver	26
engine_capacity_cc	Engine capacity	5935
age_of_vehicle	Age of vehicle	9
sex_of_driver	Driver gender code	1

The two files can be joined using the collision identifier column (collision_index).

Steps

1. Load and Inspect

Load both datasets using pandas. Display the shape, data types, and first 3 rows of each dataset. Report the number of missing values per column for both datasets.

Briefly describe what you observe about the data structure and missing values in a markdown cell.

2. Data Cleaning

The data may not be clean. Look for any inconsistencies or unusual values in the data and address them so that the data is suitable for analysis. Address whatever you find so that the data is suitable for analysis. If you see strange collision severity values, set them to NaN.

You do not need to save the cleaned data to a new file. Ensure that your cleaning steps are performed in code so that, if the notebook is run from the beginning using the original input files, the data is cleaned automatically. All subsequent analysis must be performed on the cleaned data.

Briefly explain in a markdown cell what data quality issues you found and how you addressed them.

3. Filtering and Basic Summaries

Using the cleaned collision data:

- Filter to keep only rows with a valid collision severity for use in subsequent steps.

- Display the top 10 collisions by **number_of_casualties**, showing collision_index, police_force, collision_severity, number_of_casualties, number_of_vehicles, and speed_limit.
- Display the number of collisions for each police_force (you may display the top values if the output is large)

Briefly interpret what these summaries tell you about the distribution of collisions and severity in a markdown cell.

4. Grouped Analysis

Using the filtered collision data:

- Create a new column safety_score defined as 4 minus collision_severity.
- Group by police_force and compute:
    - Average safety score (using the safety_score column)
    - Number of collisions (nunique of collision_index)
    - Total casualties (sum of number_of_casualties)

- Display numeric results with two decimal places. Sort by average safety score in descending order.
- Group by **collision_year** and **police_force** and compute:
    - Average safety score
    - Number of collisions
- Display numeric results with two decimal places.

Briefly interpret what these grouped summaries reveal about differences across police forces in a markdown cell.

5. Filters by severity and size

Using the filtered collision data:

- Identify collisions with severity less than or equal to 2 and at least 2 casualties. Report how many such collisions there are and show the first 5 rows.
- Identify collisions with severity equal to 3 and exactly 1 casualty. Report how many such collisions there are and show the first 5 rows.
- Identify collisions where collision_adjusted_severity_serious is greater than or equal to 60%. Report how many such collisions there are and show the first 5 rows.

Briefly explain what these filters reveal in a markdown cell.

6. NumPy Analysis

Using **NumPy only** (no pandas aggregation functions for this step):

- Convert speed_limit and number_of_casualties to NumPy arrays from the filtered collision data. Compute the covariance between these two variables using np.cov. Use

only rows where both values are finite. Extract the covariance between the two variables from the resulting matrix.
- Compute the 25th, 50th, and 75th percentiles of number_of_casualties using np.percentile. Use only finite values.
- Compute the proportion of collisions with at least two casualties using the mean of a boolean array.

Briefly interpret what these numerical results suggest in a markdown cell.

7. Derived Column

Create a new column **multi_vehicle_collision** in the filtered collision data.

This column should take the value 1 if number_of_vehicles is greater than or equal to 2, and 0 otherwise. Ensure that the result is a defined 0 or 1 for all rows (treat missing number_of_vehicles values as 0).

Display a sample of 5 rows showing only collision_index, number_of_vehicles, number_of_casualties, and multi_vehicle_collision.

8. Vehicle Dataset Analysis

Using the cleaned vehicle dataset:
Group by vehicle_type and compute:
- Mean age_of_driver
- Number of rows
- Number of unique collisions (using nunique)

Display numeric results with two decimal places.

9. Merge and Merged Analysis

Merge the filtered collision data with the cleaned vehicle dataset using the collision_index column.
Using the merged data, group by police_force and compute:
- Median age_of_driver (vehicle records only; coerce non-numeric values to missing)
- Standard deviation of age_of_driver
- Number of merged rows (vehicle-level records)
- Number of distinct vehicle_type values (nunique of vehicle_type within each police_force)

Display numeric results with two decimal places. Sort by median age_of_driver in descending order.
Create a pivot table with police_force as rows and vehicle_type as columns, using an aggregation that counts vehicle records in each cell (count of collision_index).
Display values with **two decimal places**.

Briefly describe what the per-cell counts and the age summaries show about involvement patterns across police force areas in a markdown cell.

10. Final Answer and Reproducibility

Using the filtered collision data at the collision level (not the merged vehicle-level rows), produce a **final analysis summary csv** that answers the following question:
Which police force has the highest average multi_vehicle_collision?

Your analysis summary csv should include:

- police_force
- average multi_vehicle_collision (Avg_Multi_Vehicle_Collision)
- number of collisions (N_Collisions)
- total casualties (Total_Casualties)

Use the names in parentheses as the column names in the output file.
Compute Avg_Multi_Vehicle_Collision as the mean of multi_vehicle_collision grouped by police_force.

In a markdown cell, state your conclusion and explain how the table supports it.

Save this summary table to:
analysis_result_road_safety_by_force.csv in **datasets/** folder.

Your notebook must be runnable from top to bottom and must produce this file. Do not submit the input CSV or this final CSV file separately; it will be regenerated when your notebook is executed for marking.

Submission

Your analysis notebook

- File name: assignment2_<yourID>.ipynb
    - Must include:
        - Code for all analysis tasks
        - Short written explanations after each task as requested above

Submit your Jupyter Notebook only (**.ipynb**). **Do NOT submit the datasets, the output CSV, or any other files.**

Good luck!